

Inverse variation



1 $y = \frac{k}{x}$

2 $k = 150$

3 a Direct

b Inverse

4 $r = \frac{k}{a}$

5 a No
e Yes

b No
f No

c No

d Yes

6 a 375

b $\frac{125}{2}$

c $\frac{125}{4}$

7 a Yes

b Yes

c No

d No

8 a $k = 252$

b $x = 9, y = 42$

9 a $t = \frac{48}{s}$

b $t = \frac{2}{s^2}$

10 a 0

b 0

Key features of hyperbola

11 a 0
e $y = x, y = -x$

b ∞

c $x = 0$

d $y = 0$

12



13

a

x	-2	-1	$-\frac{1}{2}$	$-\frac{1}{10}$	$-\frac{1}{100}$	$\frac{1}{100}$	$\frac{1}{10}$	$\frac{1}{2}$	1
y	-1	-2	-4	-20	-200	200	20	4	2

b 0

c $x = \frac{2}{y}$

d 0

e As x approaches 0 from the right, the function value approaches ∞ . As x approaches 0 from the left, the function value approaches $-\infty$.
As x approaches ∞ and $-\infty$, y approaches 0. This is called the limiting value of the function.

14

a 0

b $-\infty$ c ∞

d 0

e 0

15 6

16

a Yes

b No

c No

17

a $x = 0$ b $y = 0$ c All real values of x , except $x = 0$.d All real values of y , except $y = 0$.

18 The function will be undefined.

19

a $x = 0$ only.b All real values of x , except $x = 0$.c $y = 0$ d All real values of y , except $y = 0$.

20 Looking at $y = \frac{2}{x}$, notice that the numerator 2 is non-zero, and so $\frac{2}{x}$ can never be equal to 0. Another approach is to rearrange the equation and make x the subject. Then we get $x = \frac{2}{y}$. In this form we can see that the denominator y cannot be equal to 0.

21

a 2

c $y = \frac{4}{x}$



b 1

d The graph moves further from the axes

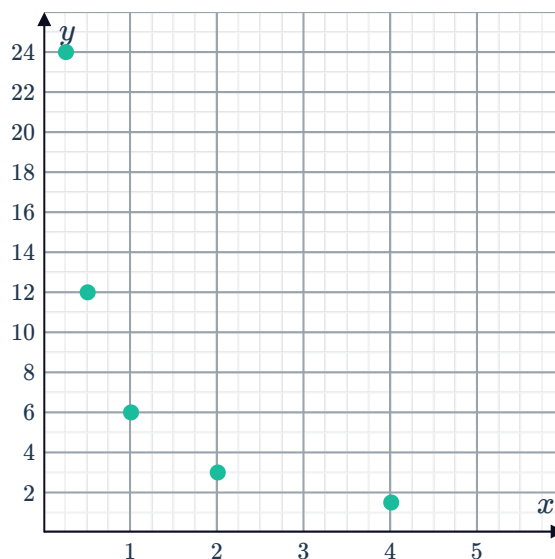
Graphs of hyperbolas

22

a

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4
y	24	12	6	3	$\frac{3}{2}$

b



23 She can label a point on the graph.

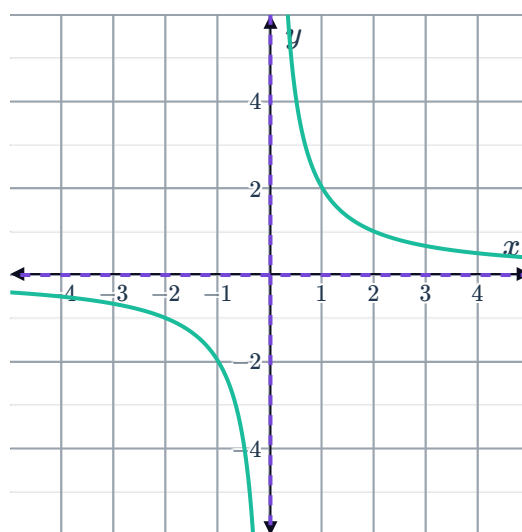
24

a

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	-1	-2	-4	4	2	1

ii



iii 1 and 3

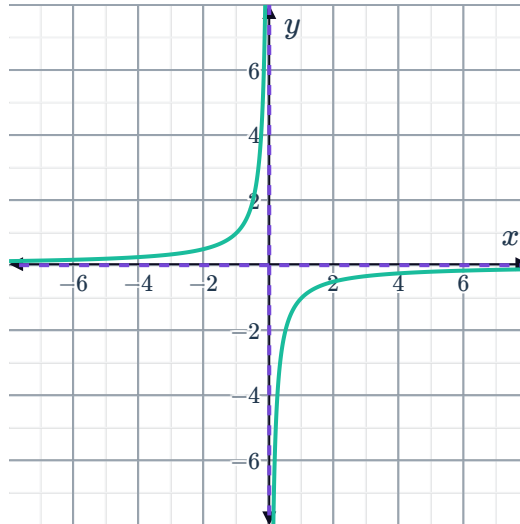
b

i



x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{2}$	1	2	-2	-1	$-\frac{1}{2}$

ii



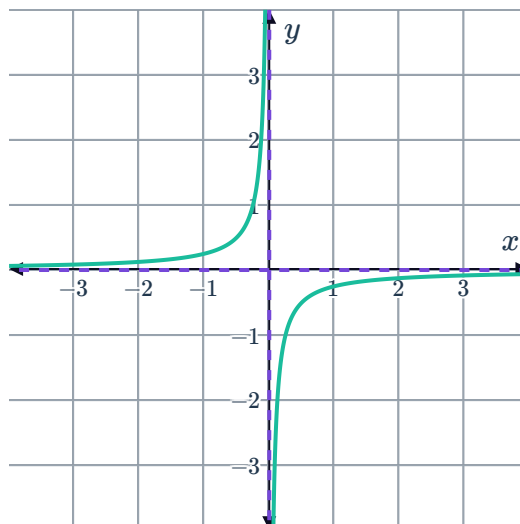
iii 2 and 4

c

i

x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{4}$	$-\frac{1}{8}$

ii



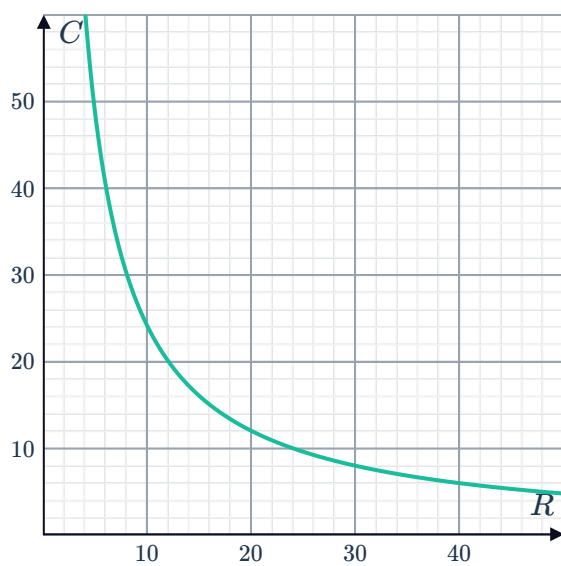
iii 2 and 4

Applications



25

a



b The current decreases.

26

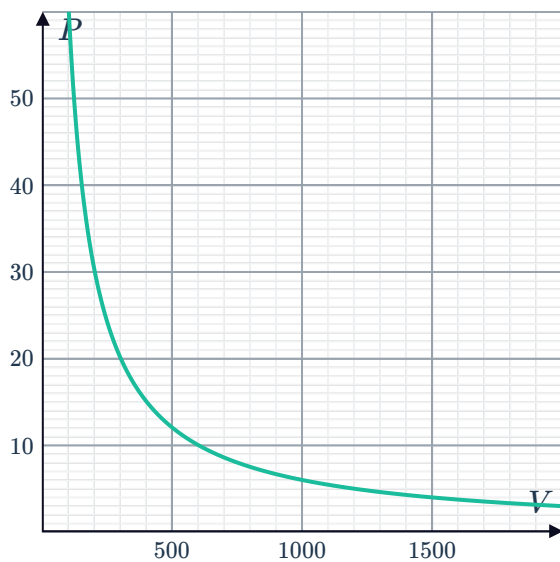
a

-2

b 32 units^2

27

a

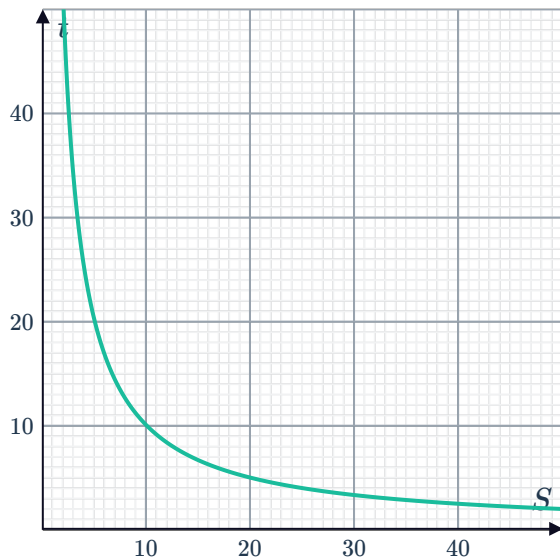


b 6000 kg/cm^2

c The pressure decreases.

28

a



b 10 hours

c 2 hours

d The speed must increase.